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import pandas as pd

import nltk

from nltk.sentiment import SentimentIntensityAnalyzer

from textblob import TextBlob

from nrclex import NRCLex

from sklearn.metrics import classification_report

import matplotlib.pyplot as plt


nltk.download('vader_lexicon')


df = pd.read_csv('sentiment_dataset.csv')

df['date'] = pd.to_datetime(df['date'])


# VADER

sia = SentimentIntensityAnalyzer()

df['vader_score'] = df['text'].apply(lambda x: sia.polarity_scores(x)['compound'])

df['vader_sent'] = df['vader_score'].apply(lambda c: 'Positive' if c>=0.05 else ('Negative' if
c<=-0.05 else 'Neutral'))


# TextBlob

df['blob_polarity'] = df['text'].apply(lambda x: TextBlob(x).sentiment.polarity)

df['blob_sent'] = df['blob_polarity'].apply(lambda p: 'Positive' if p>0 else ('Negative' if p<0 else
'Neutral'))


# NRC Emotion

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def get_emotion(text):
    try:
        e = NRCLex(text).affect_frequencies
        return max(e, key=e.get) if e else 'neutral'
    except:
        return 'unknown'

df['nrc_emo'] = df['text'].apply(get_emotion)


# Evaluation

print("=== VADER vs Original Sentiment ===")
print(classification_report(df['sentiment'], df['vader_sent']))


print("\n=== NRC vs Original Emotion ===")
print(classification_report(df['emotion'], df['nrc_emo']))


# Trend plot

df['month'] = df['date'].dt.to_period('M')
trend = df.groupby('month')['sentiment'].value_counts().unstack().fillna(0)
trend.plot(kind='line', marker='o')
plt.title('Sentiment Trend Over Time')
plt.ylabel('Count')
plt.savefig('sentiment_trend.png')
plt.show()

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# Save enriched data

df.to_csv('sentiment_analyzed_output.csv', index=False)

print("Output saved to sentiment_analyzed_output.csv")
```